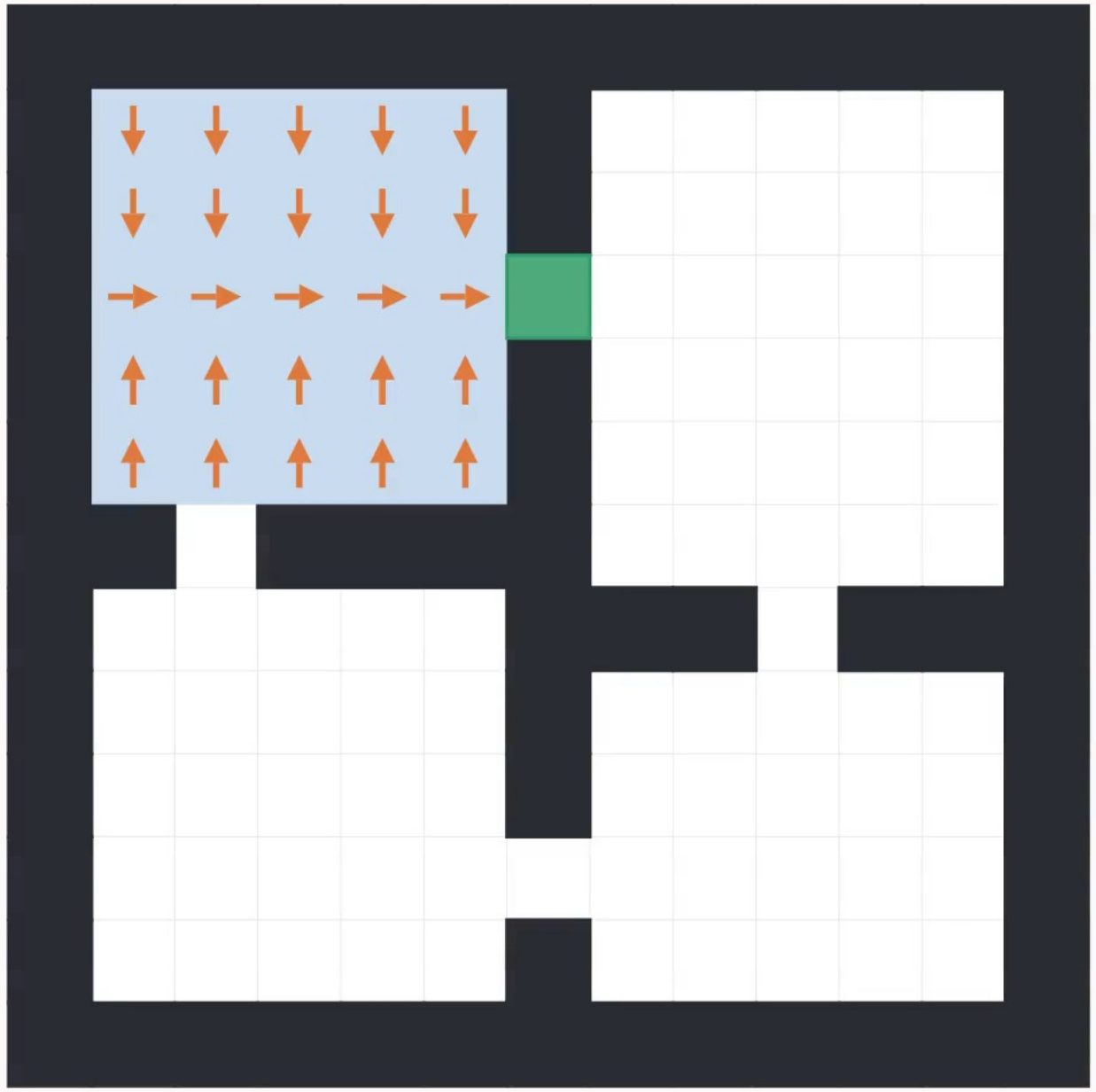
So how do we use a set of options?

1. **Stitch**
   learn a policy over options: pick one, run it, pick the next
2. **Combine**
   blend options with GPE + GPI into a new one, no extra learning
3. **Both: the Option Keyboard**
   learn a policy over chords; each chord makes a new option
4. **a secret 4th thing**

Goal-Space Planning

Anastasiia Pedan



What is an option?

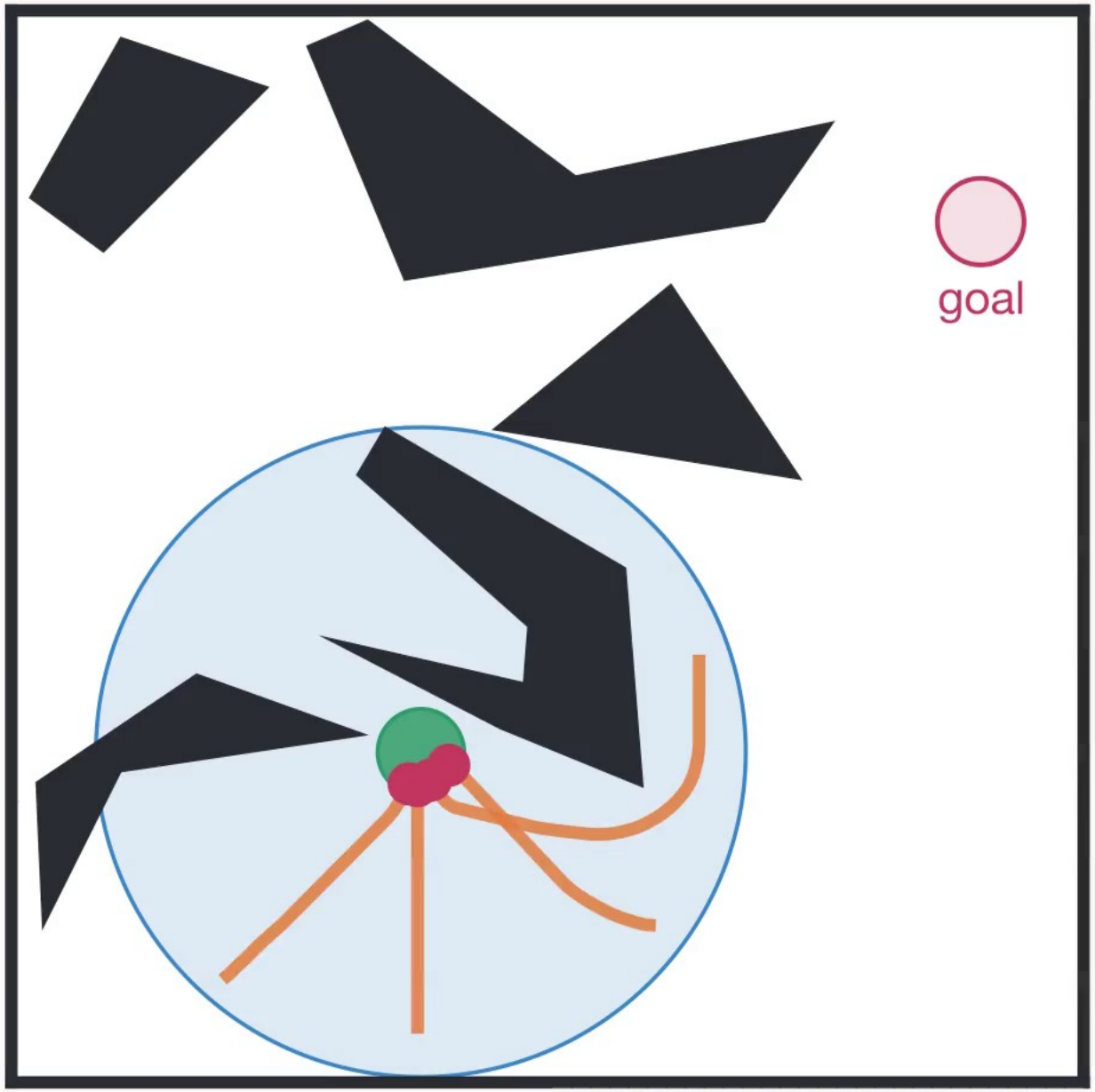
$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

"go to the hallway"

$\mathcal{I}$ where it can start

π how it acts

β where it stops



Options in Pinball

$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

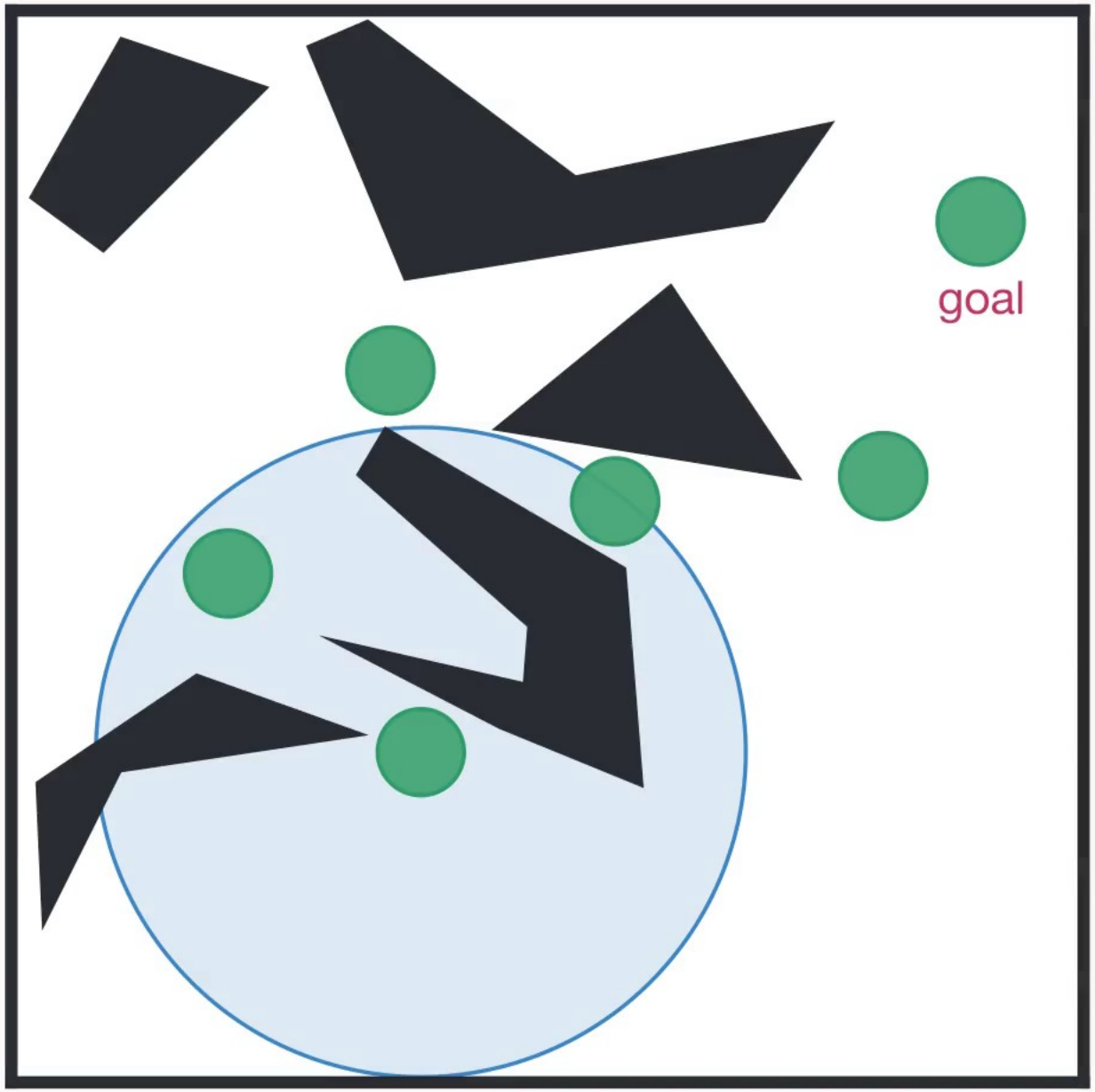
"go to the subgoal"

$\mathcal{I}$ where it can start

π how it acts

β where it stops

one option, many starting states



Options in Pinball

$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

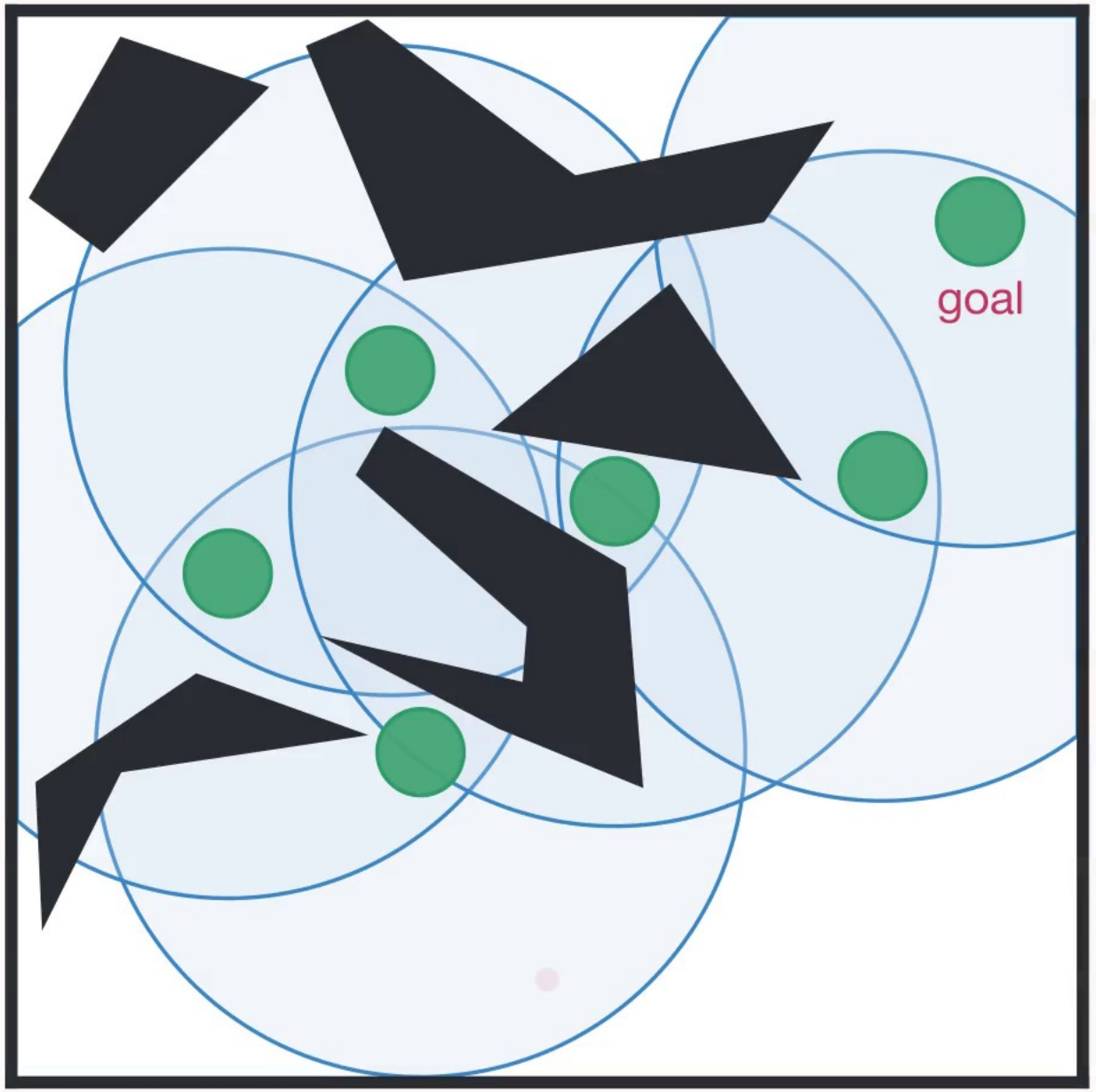
"go to the subgoal"

$\mathcal{I}$ where it can start

π how it acts

β where it stops

add more subgoals



Options in Pinball

$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

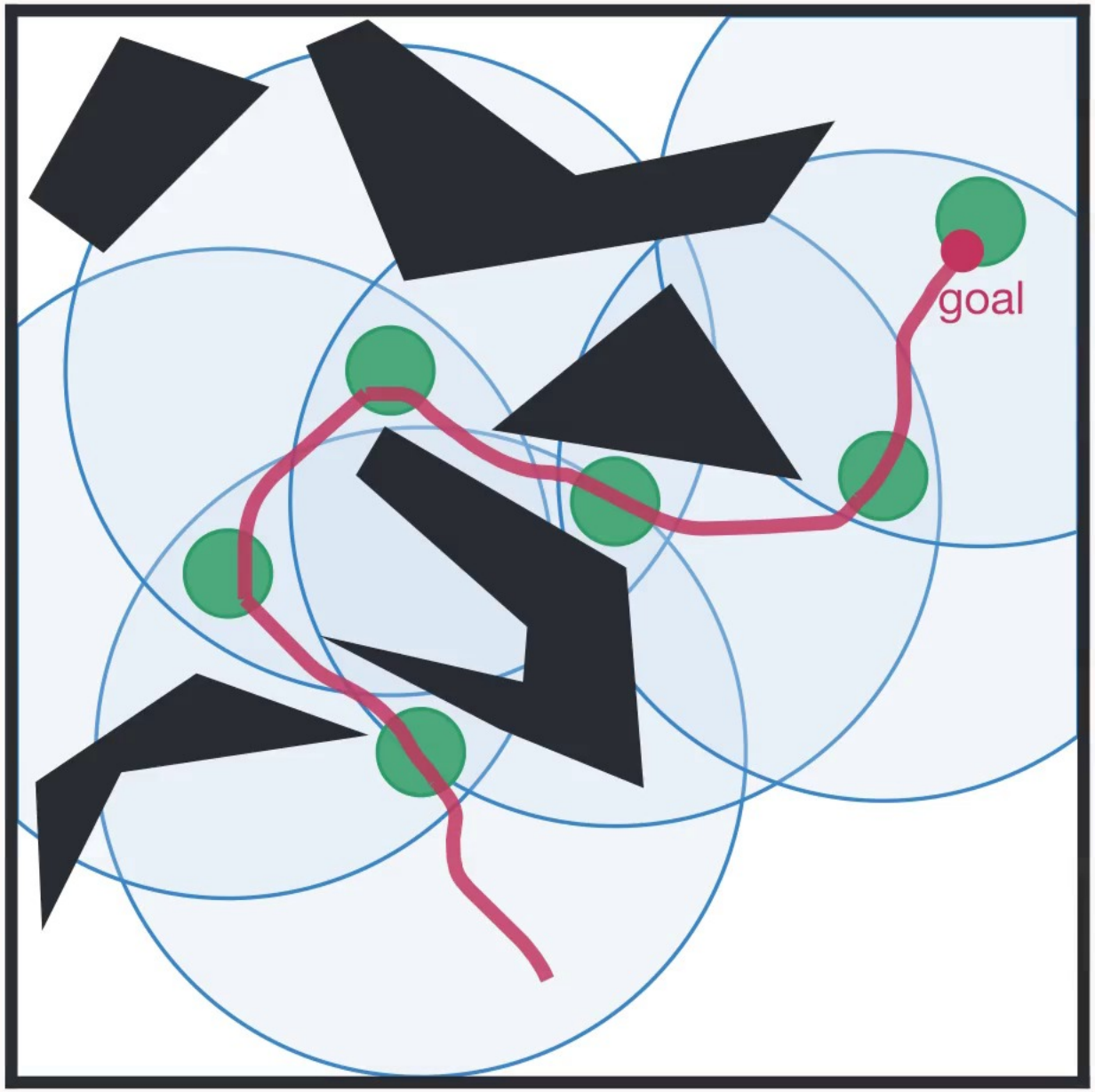
"go to the subgoal"

$\mathcal{I}$ where it can start

π how it acts

β where it stops

each subgoal lies in the next option's $\mathcal{I}$



Options in Pinball

$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

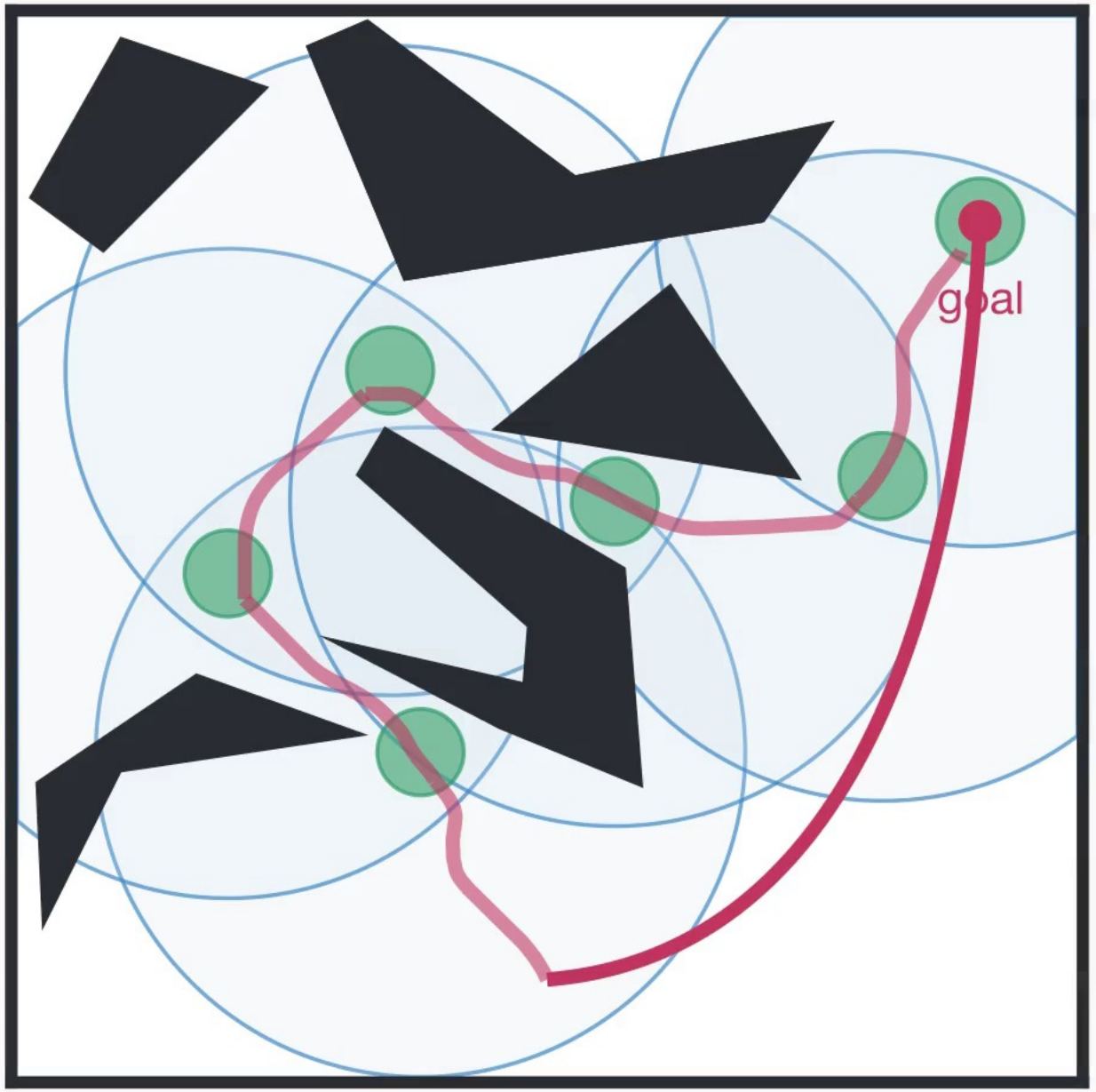
"go to the subgoal"

$\mathcal{I}$ where it can start

π how it acts

β where it stops

stitch options to reach the goal



Options in Pinball

$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

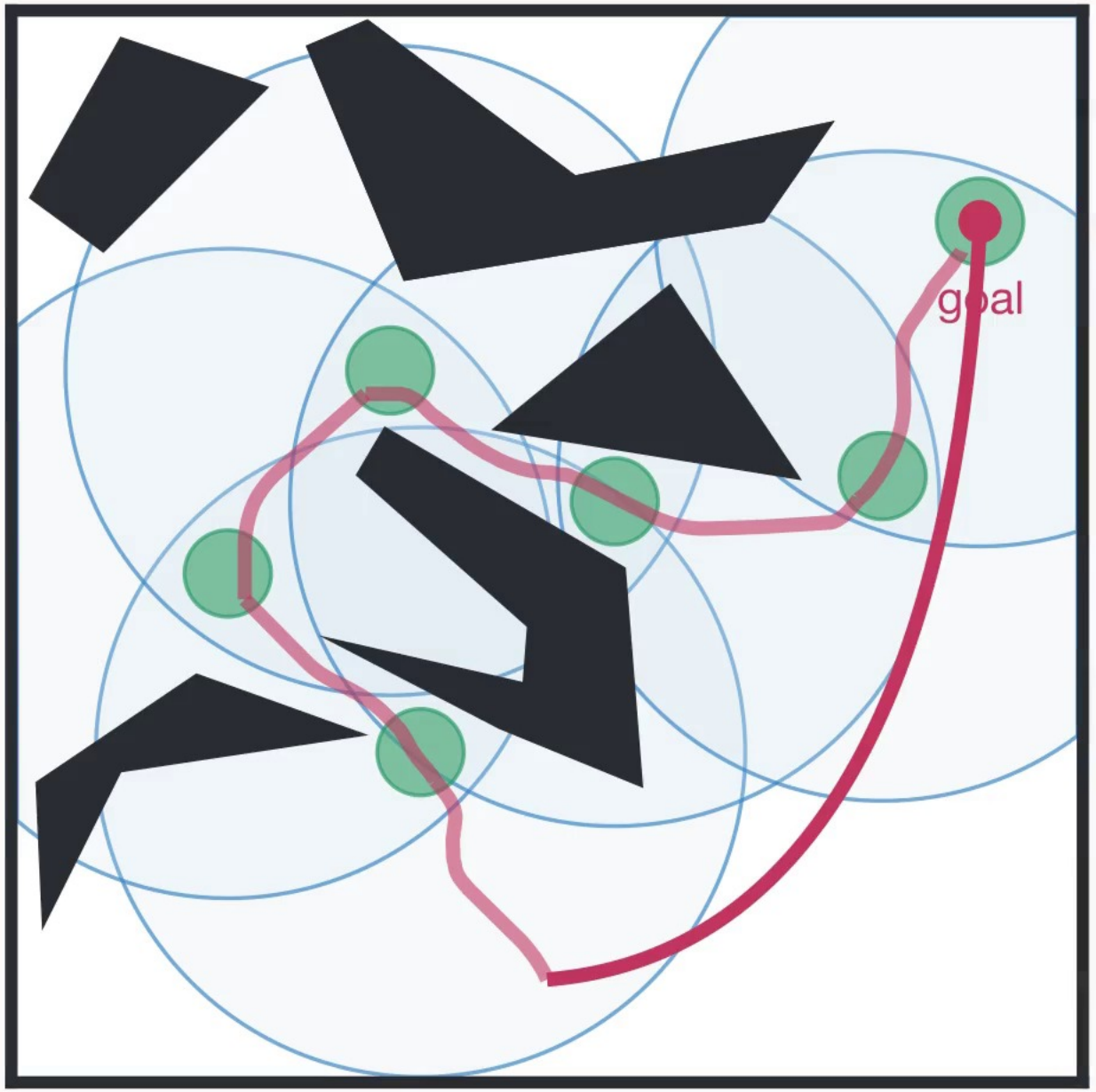
"go to the subgoal"

$\mathcal{I}$ where it can start

π how it acts

β where it stops

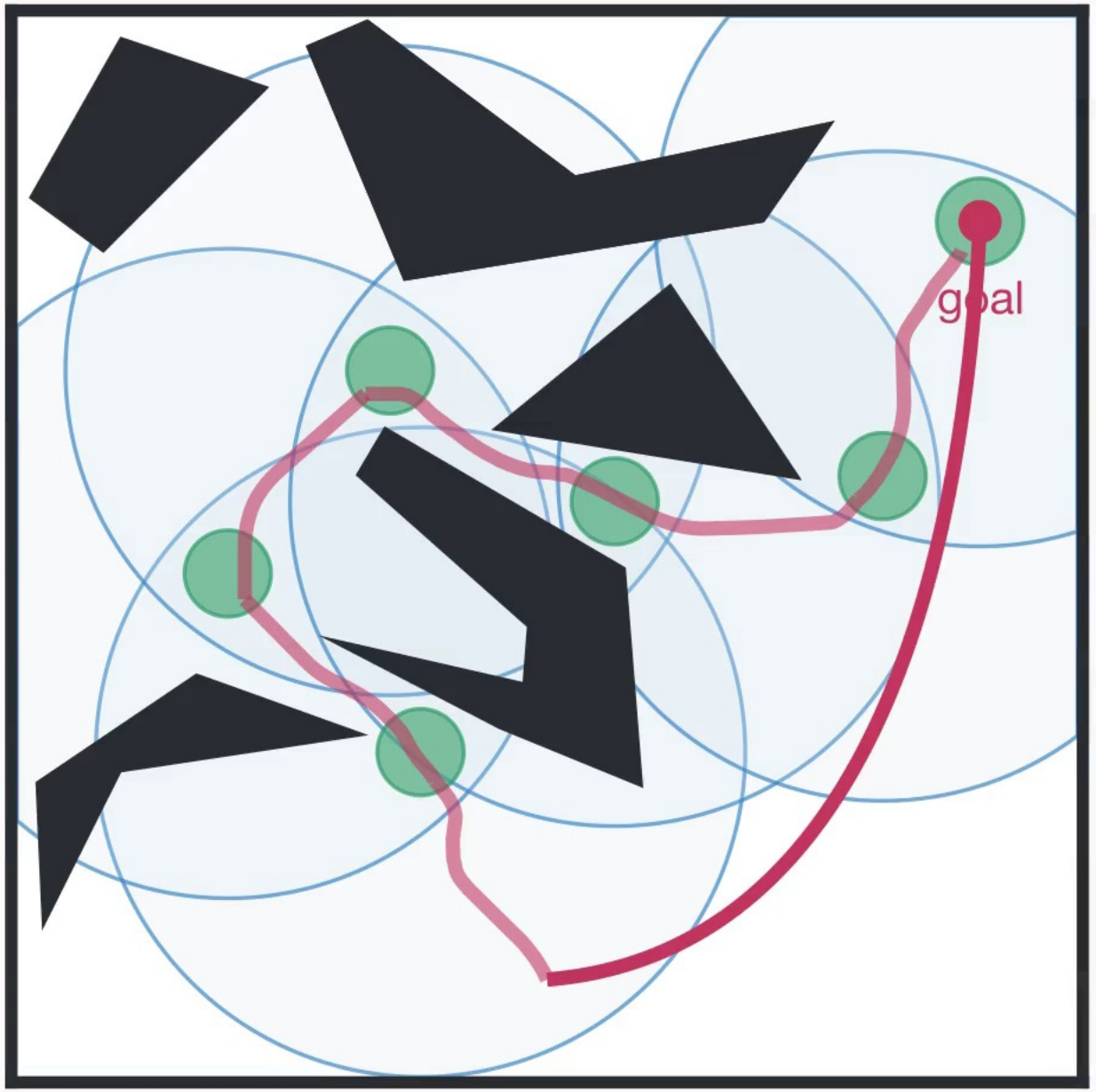
But this shortcut is faster



Options in Pinball

Can we still use the options even if executing them is suboptimal?

We can learn *about* them!



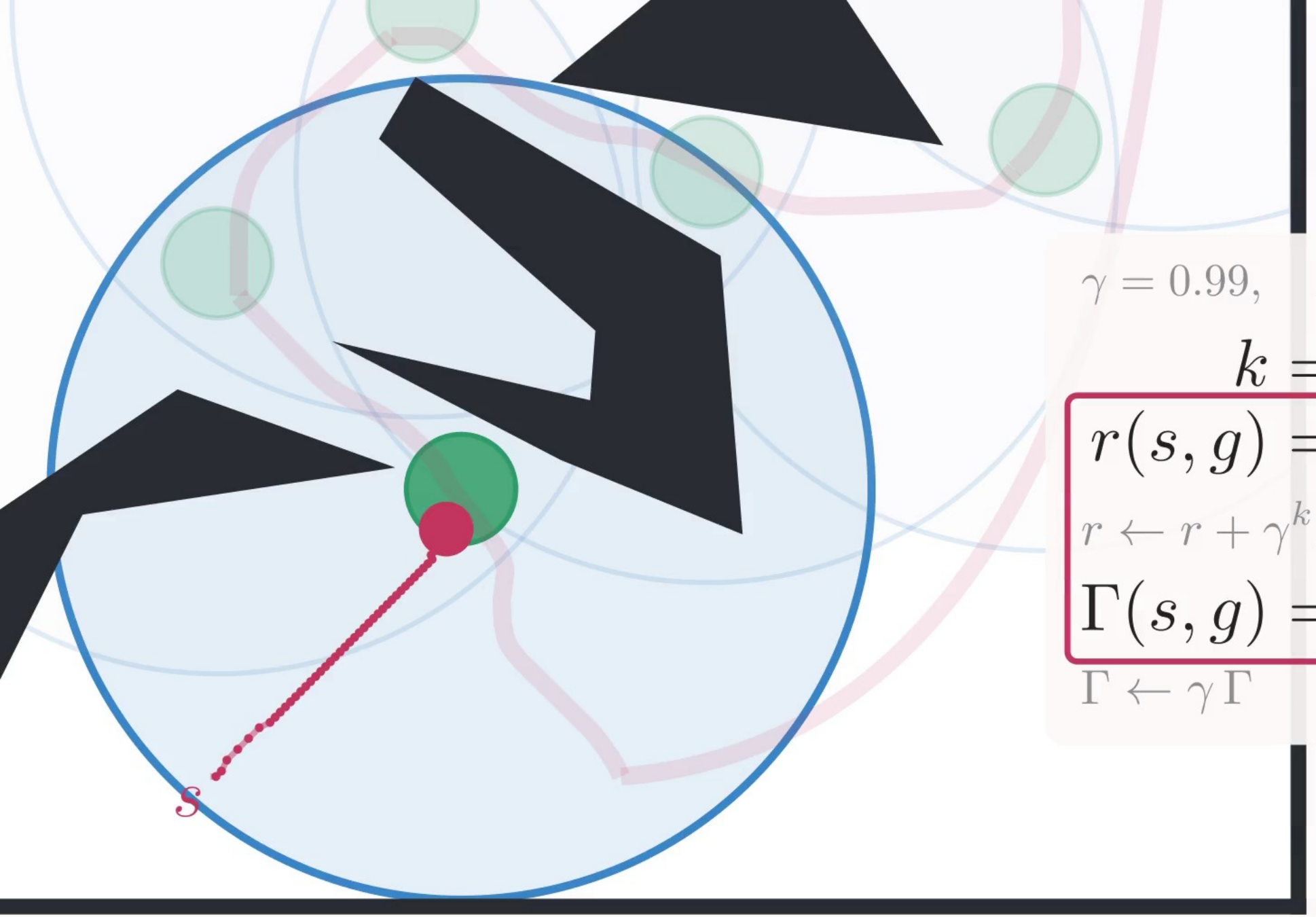
Subgoal models

$$r(s,g) = \sum_{k=0}^{T-1} \gamma^k \, R_{t+k+1}$$

discounted return along g's option from s

$$\Gamma(s,g) = \begin{cases} \gamma & \text{if } s' \in g \\ \gamma \, \Gamma(s',g) & \text{otherwise} \end{cases}$$

discount accumulated until it reaches g



$\gamma = 0.99, \quad R = -1$ per step

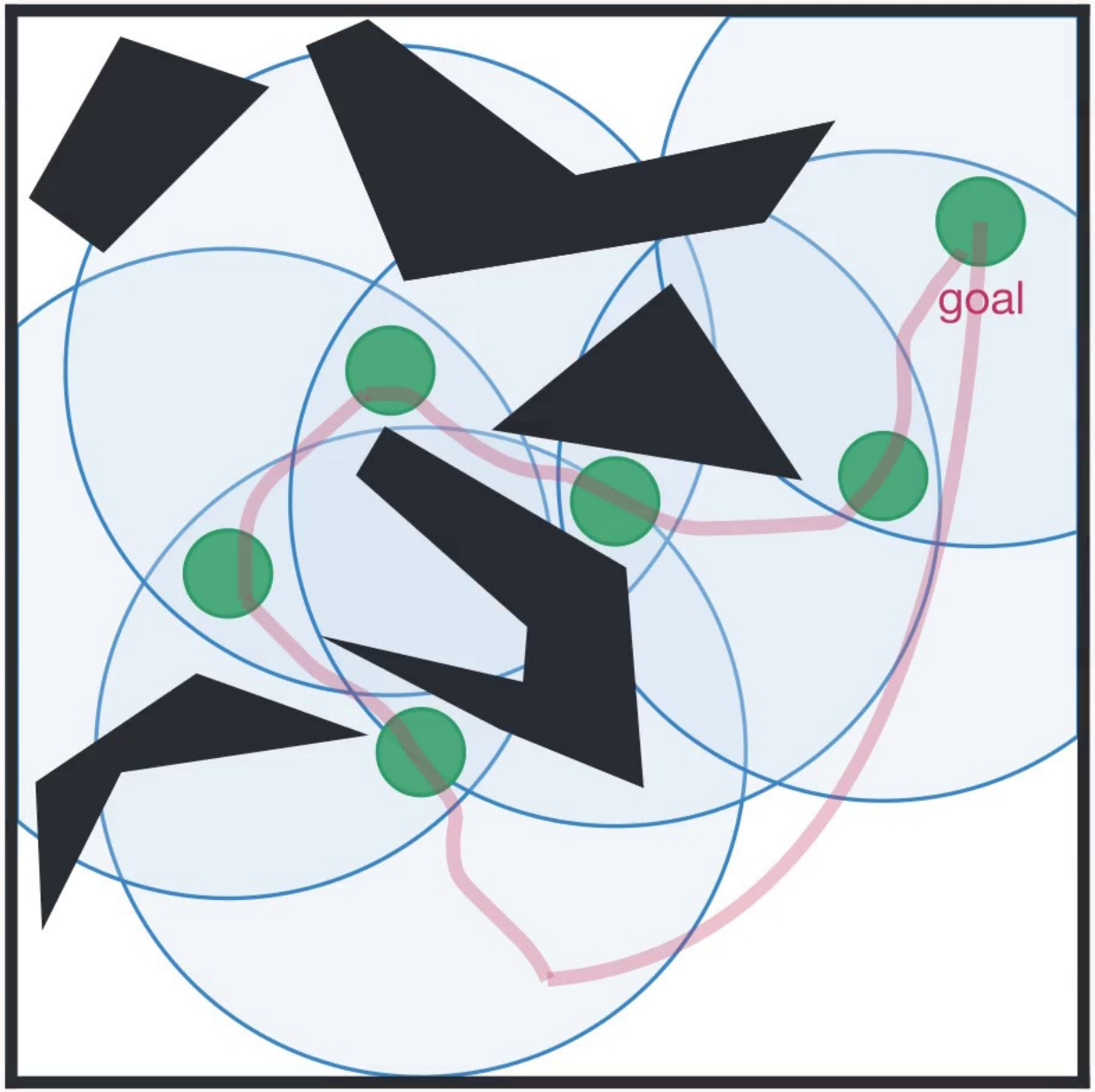
$$k = 48$$

$$r(s, g) = -38.27$$

$r \leftarrow r + \gamma^k R_{k+1}$

$$\Gamma(s, g) = 0.617$$

$\Gamma \leftarrow \gamma \Gamma$



Subgoal models

$$r(s,g) = \sum_{k=0}^{T-1} \gamma^k \, R_{t+k+1}$$

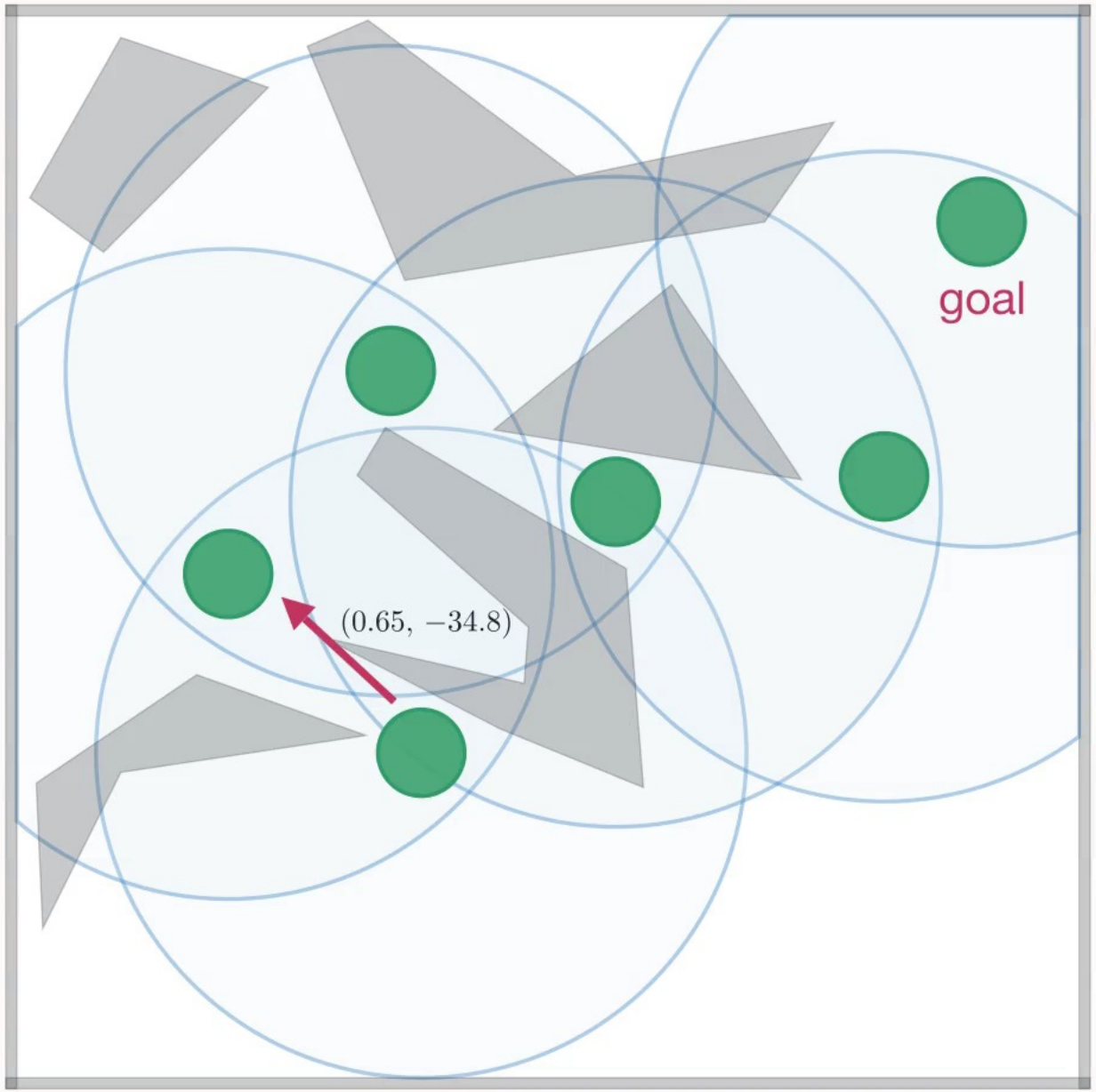
discounted return along g's option from s

$$\Gamma(s,g) = \begin{cases} \gamma & \text{if } s' \in g \\ \gamma \, \Gamma(s',g) & \text{otherwise} \end{cases}$$

discount accumulated until it reaches g

- local: only defined on each subgoal's initiation set $\mathcal{I}$
- predict scalars (r, Γ), not next-state vectors

easier to learn than dynamics models



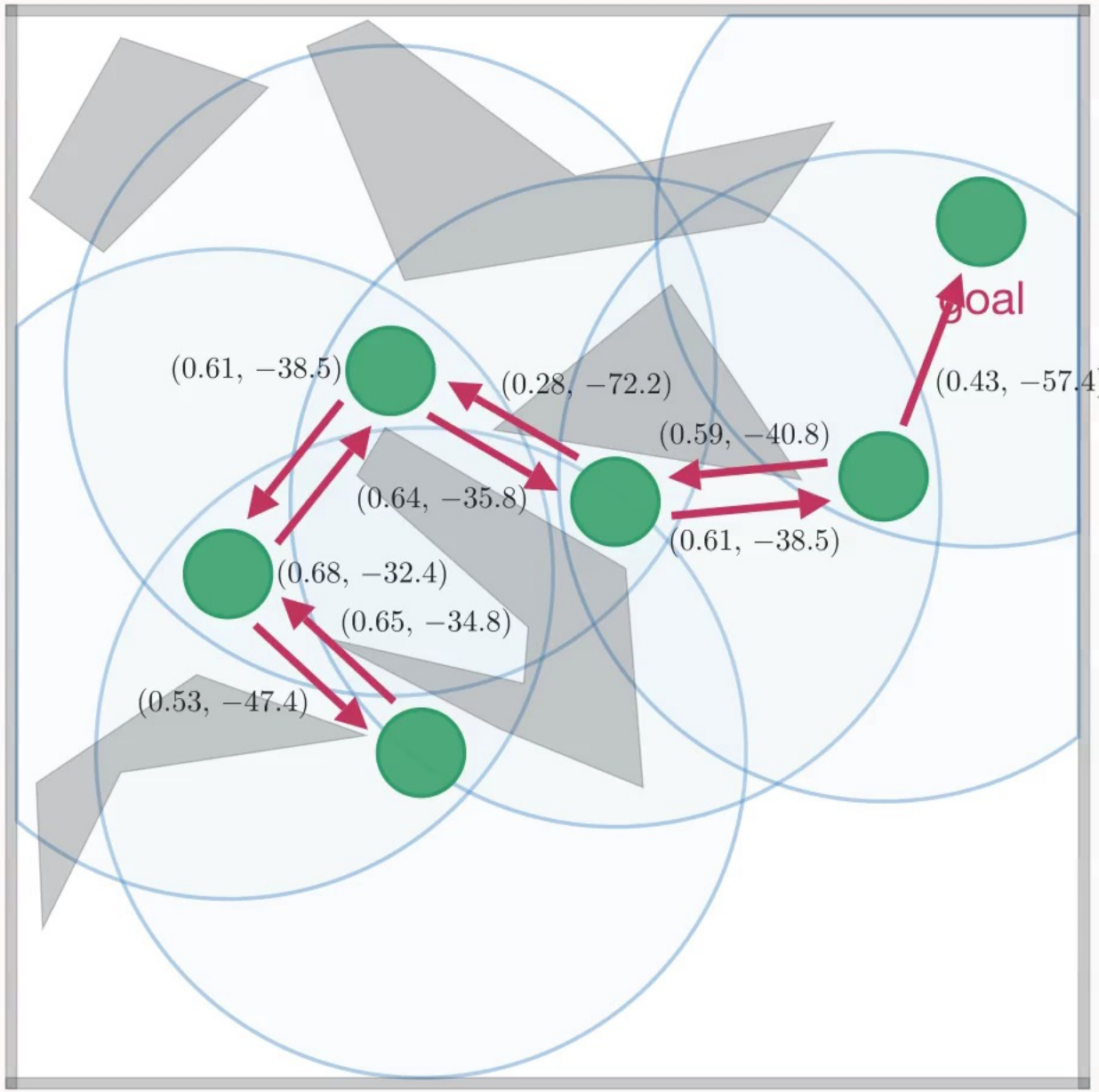
Abstract graph

use the subgoal models to build a temporally-abstract graph

$$\bar{\Gamma}(g, g') = \frac{1}{N} \sum_{i=1}^{N} \Gamma(s_i, g') = 0.65$$

$$\bar{r}(g, g') = \frac{1}{N} \sum_{i=1}^{N} r(s_i, g') = -34.8$$

s_i sampled in g's termination region



Abstract graph

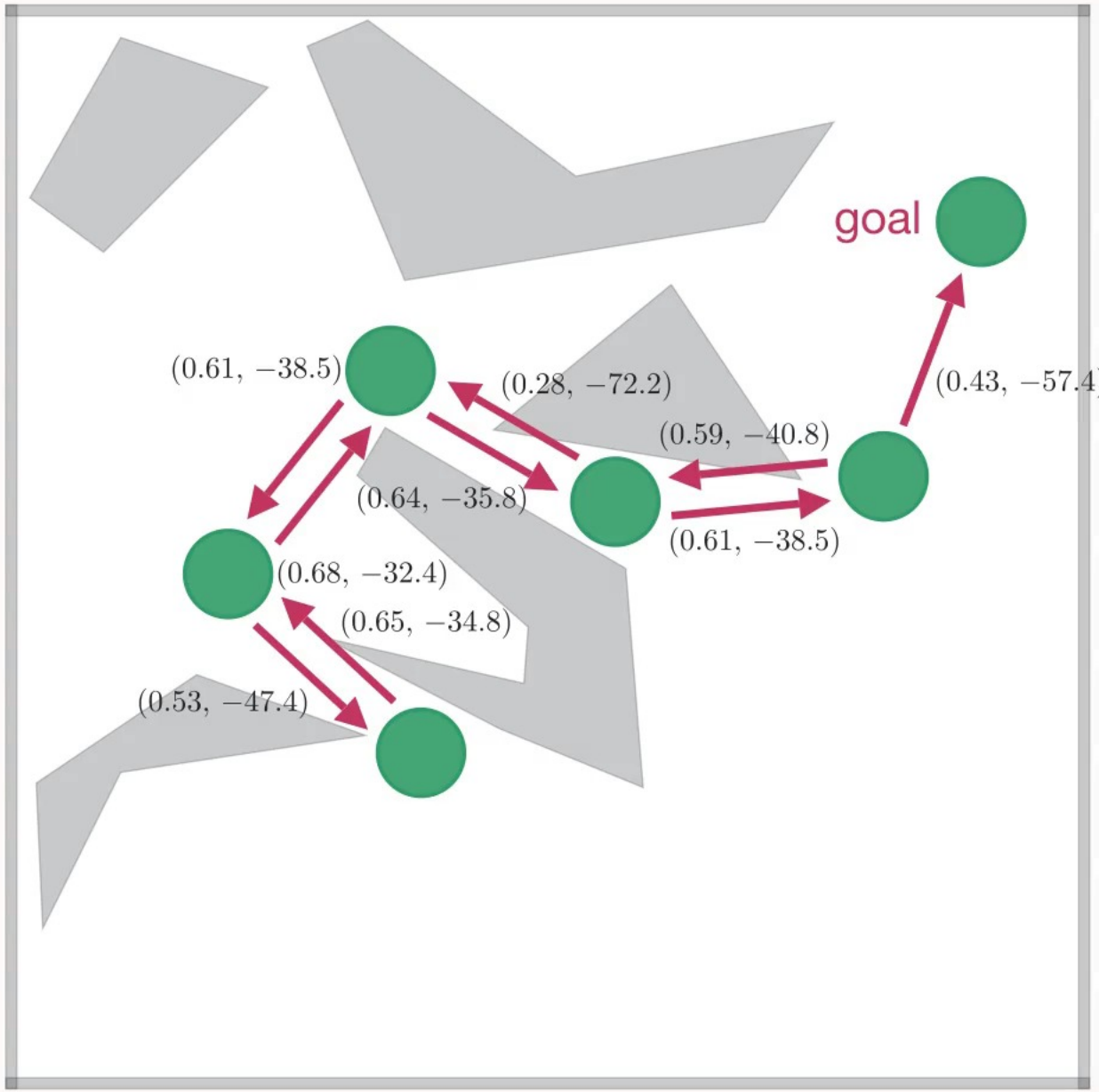
use the subgoal models to build a temporally-abstract graph

$$\bar{\Gamma}(g, g') = \frac{1}{N} \sum_{i=1}^{N} \Gamma(s_i, g') = 0.65$$

$$\bar{r}(g, g') = \frac{1}{N} \sum_{i=1}^{N} r(s_i, g') = -34.8$$

s_i sampled in g's termination region

repeat for every pair of neighbouring subgoals



Abstract graph

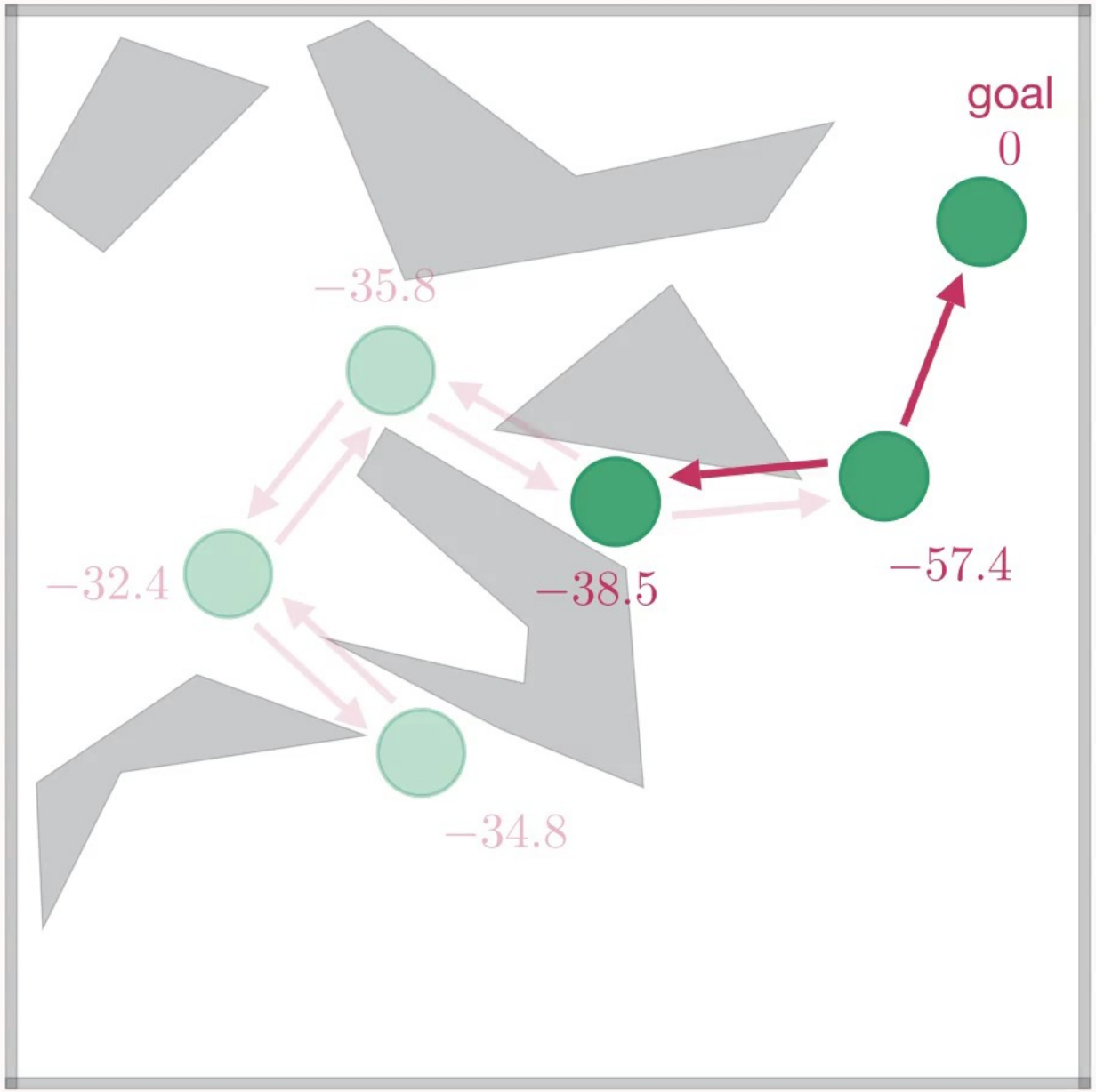
use the subgoal models to build a temporally-abstract graph

$\bar{\Gamma}$ transition probability

$\bar{r}$ transition reward

edge label: $(\bar{\Gamma}, \bar{r})$

the abstract environment representation



Value iteration

$$v^*(g) = \max_{g'} \left[\bar{r}(g, g') + \bar{\Gamma}(g, g')\, v^*(g') \right]$$

approximate value of each subgoal

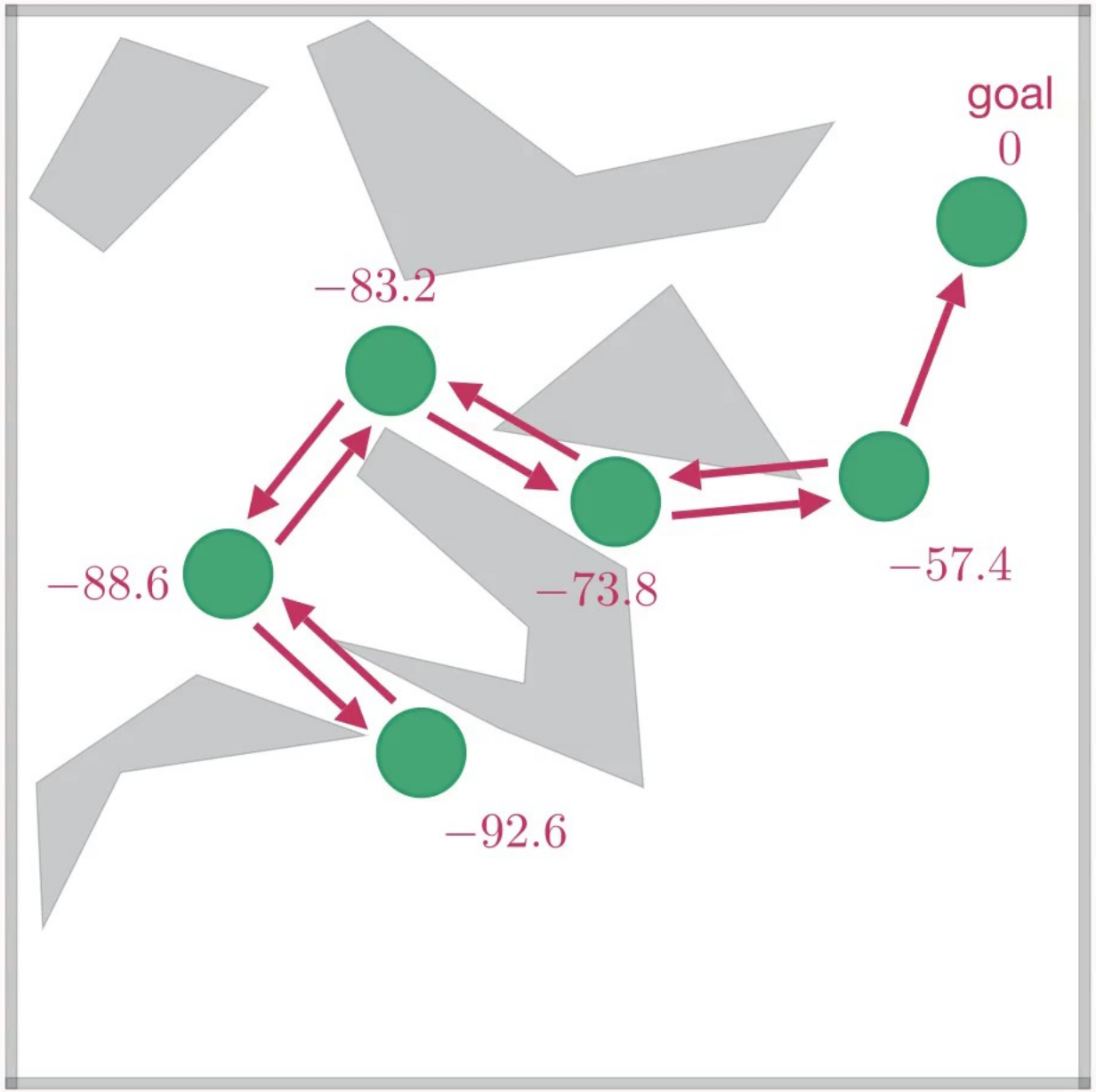
$v^*(\text{goal}) = 0$ (the episode ends there)

sweep 2

to back: $-40.8 + 0.59 \cdot (-38.5) = -63.6$

to goal: $-57.4 + 0.43 \cdot 0 = -57.4$

$\max \Rightarrow -57.4$



Value iteration

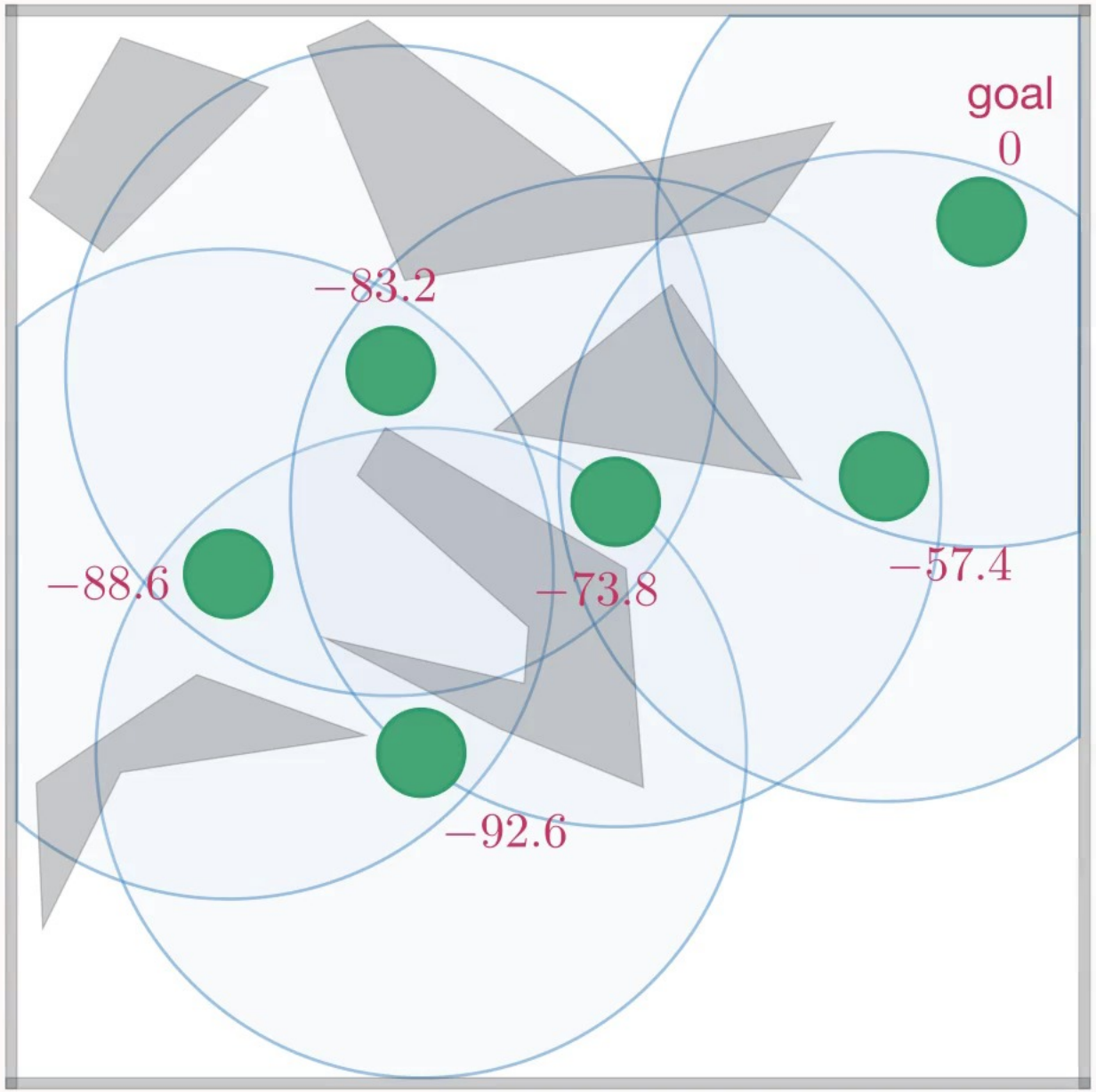
$$v^*(g) = \max_{g'} \left[\bar{r}(g, g') + \bar{\Gamma}(g, g')\, v^*(g') \right]$$

approximate value of each subgoal

$v^*(\text{goal}) = 0$ (the episode ends there)

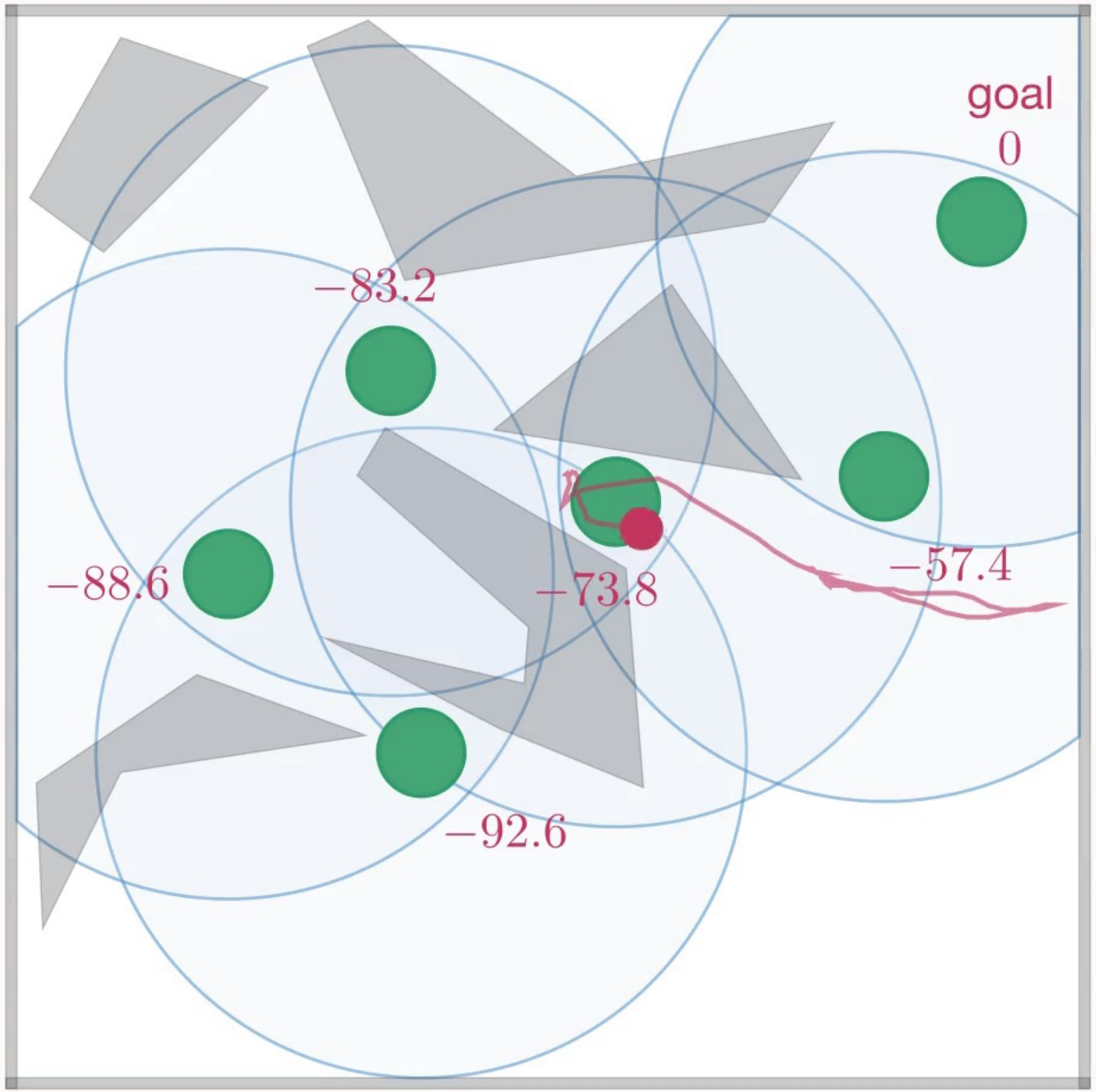
sweep 8

converged: v* for every subgoal



Back to states

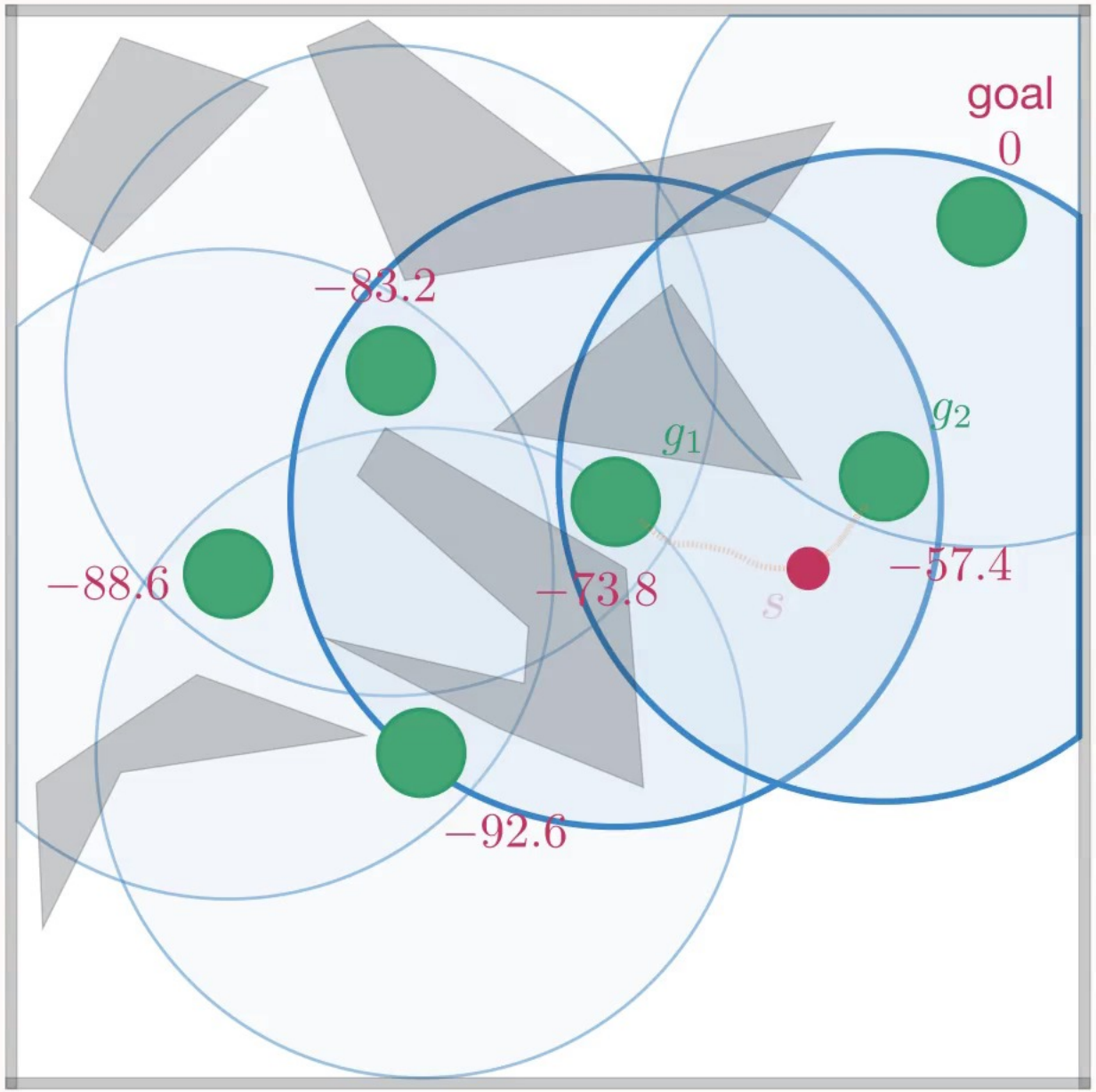
But our agent still acts in the original state-action space: how do we use these values?



Back to states

random actions, R = −1 per step

return -76



Back to states

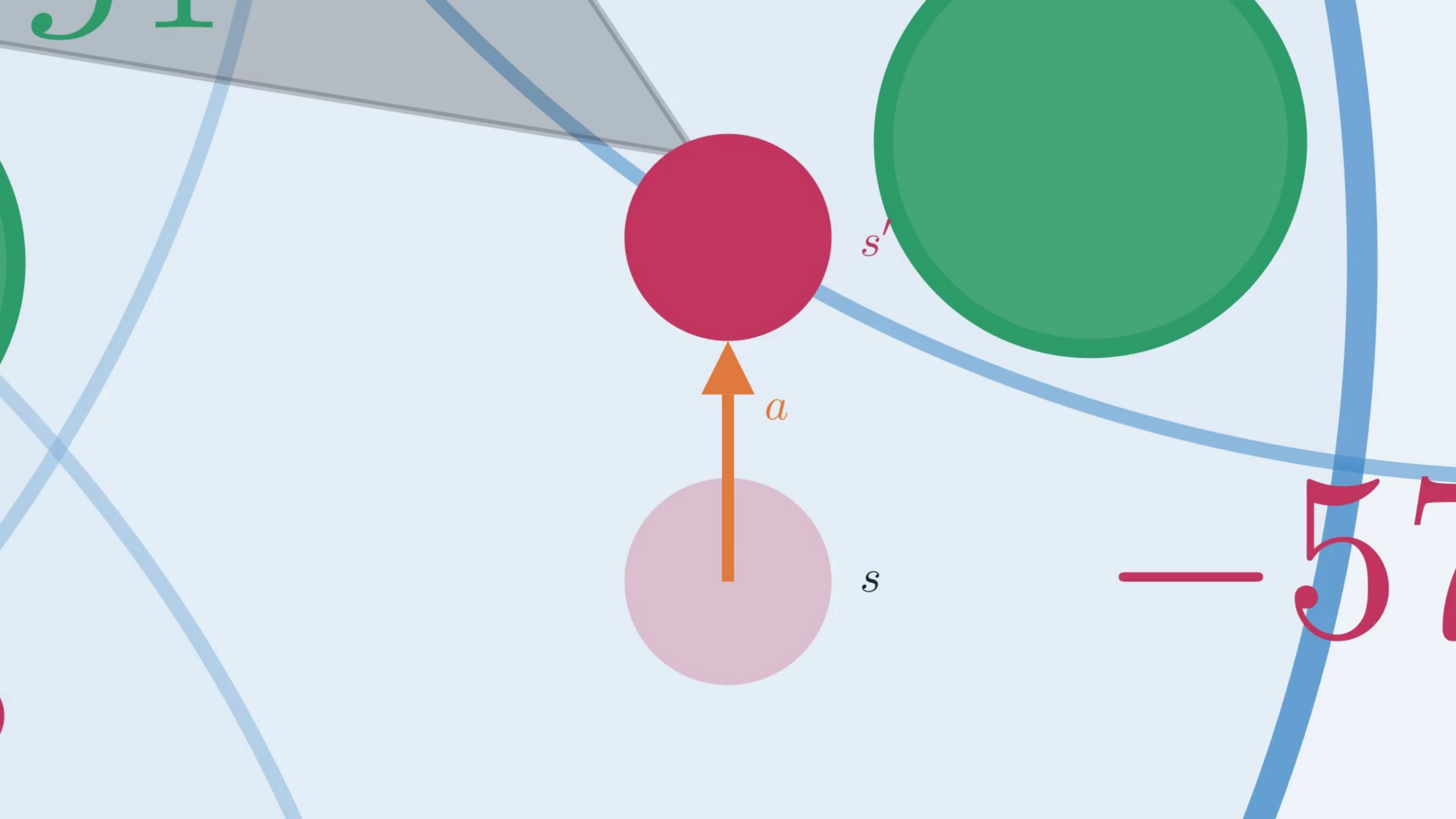
$$v_p(s) = \max_g \left[r(s,g) + \Gamma(s,g)\, v^*(g) \right]$$

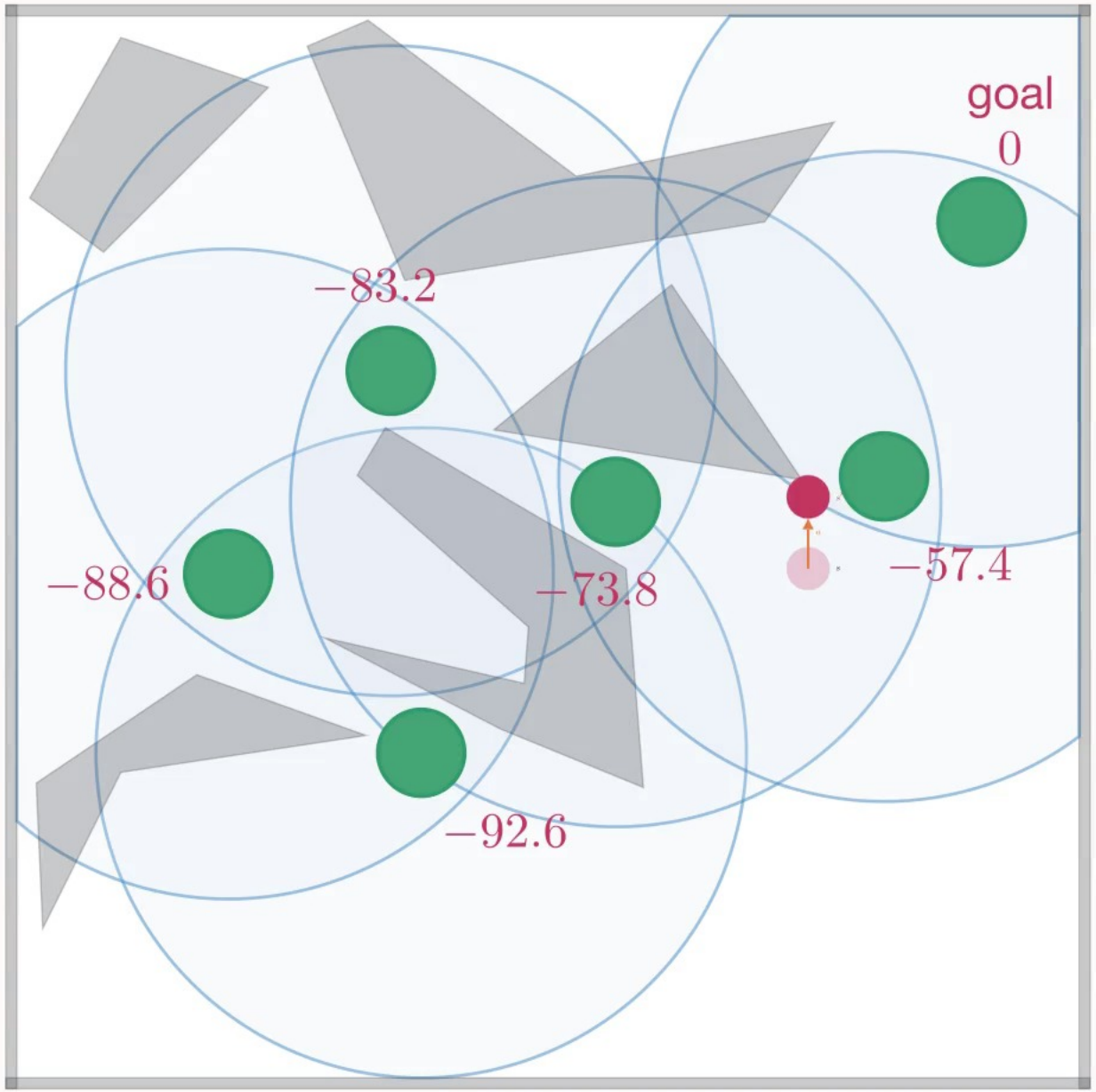
over the subgoals whose initiation set contains s

$$g_1 : \; -32.4 + 0.68 \cdot (-73.8) = -82.30$$

$$\boxed{g_2 : \; -14.9 + 0.85 \cdot (-57.4) = -63.72}$$

$$v_p(s) = -63.72$$


s'
a
s



Back to states

$$v_p(s) = \max_g \left[r(s,g) + \Gamma(s,g)\, v^*(g) \right]$$

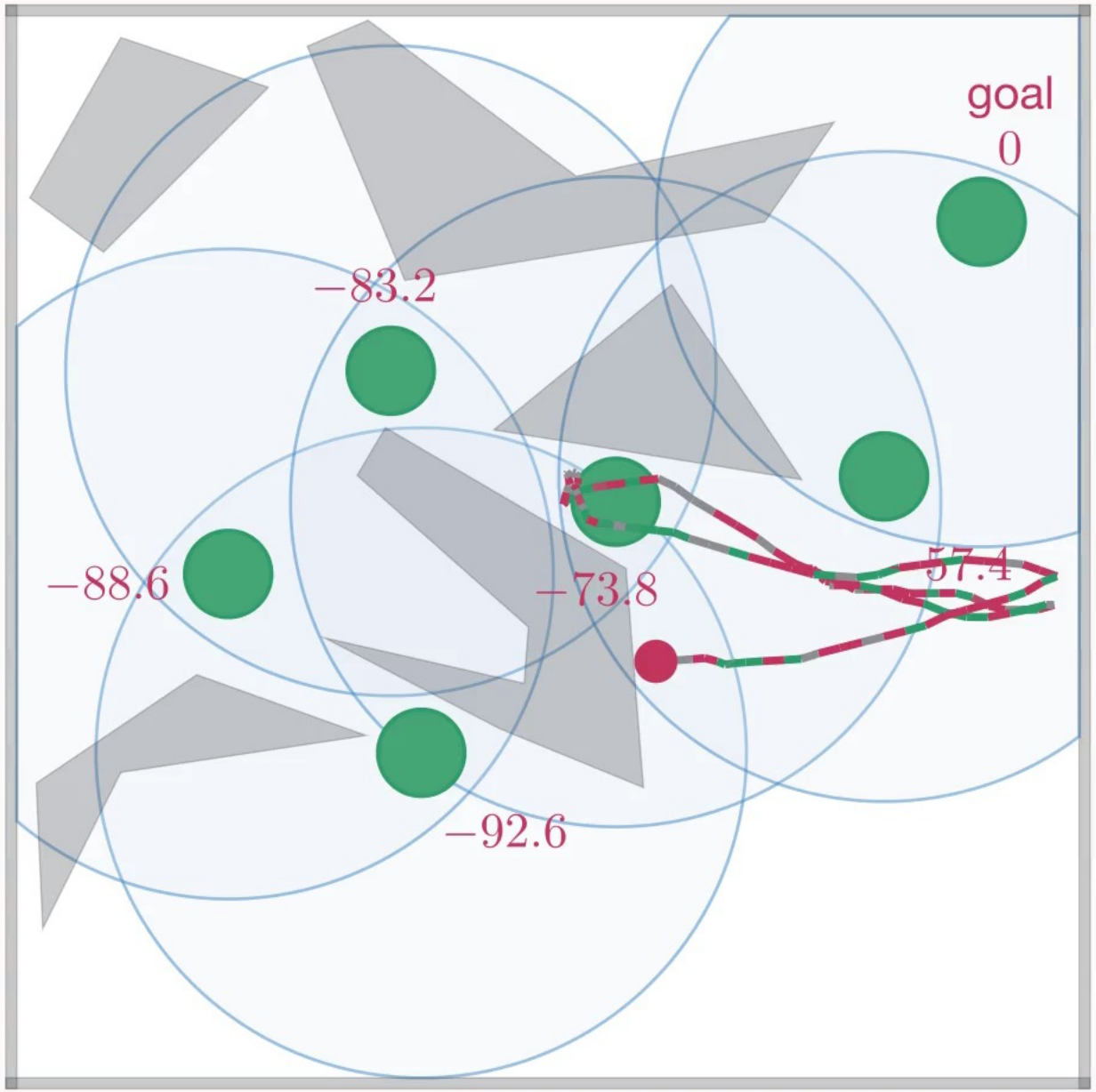
$v_p(s) = -63.72 \qquad v_p(s') = -62.98$

$$F(s,s') = \gamma\, v_p(s') - v_p(s)$$

$$= 0.99 \cdot (-62.98) - (-63.72) = 1.37$$

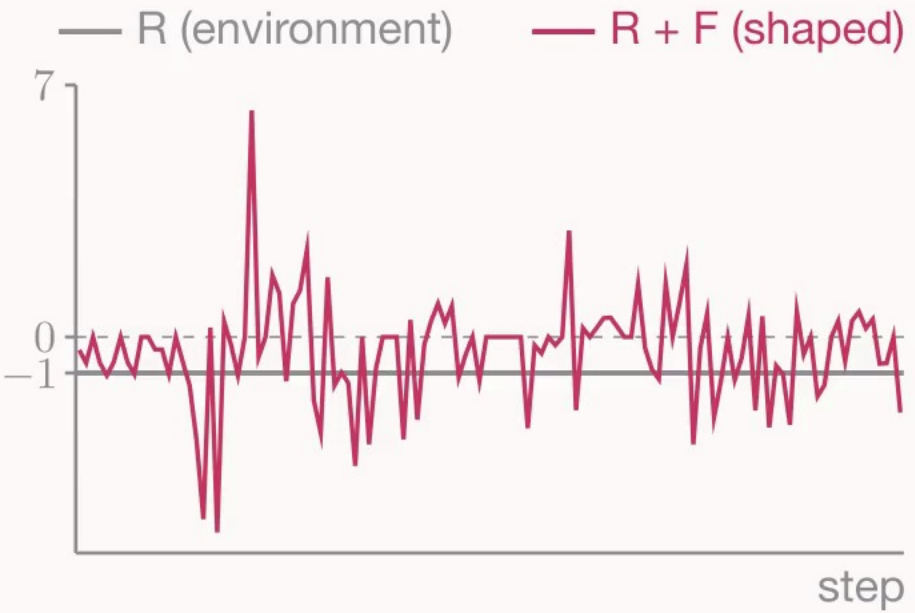
$R + F = -1 + 1.37 = +0.37$

added to the environment's reward

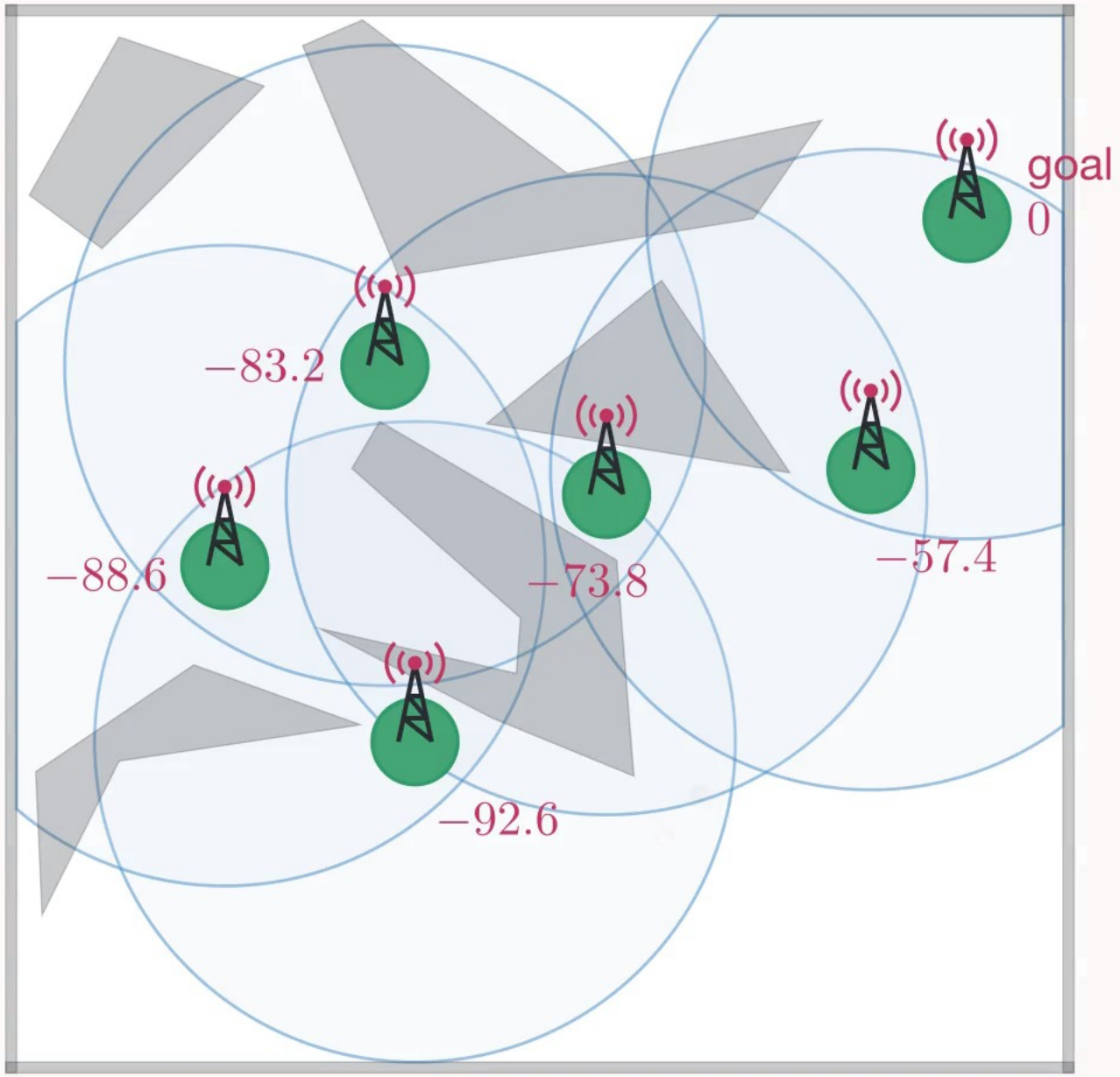


Back to states

$$F(s, s') = \gamma\, v_p(s') - v_p(s)$$



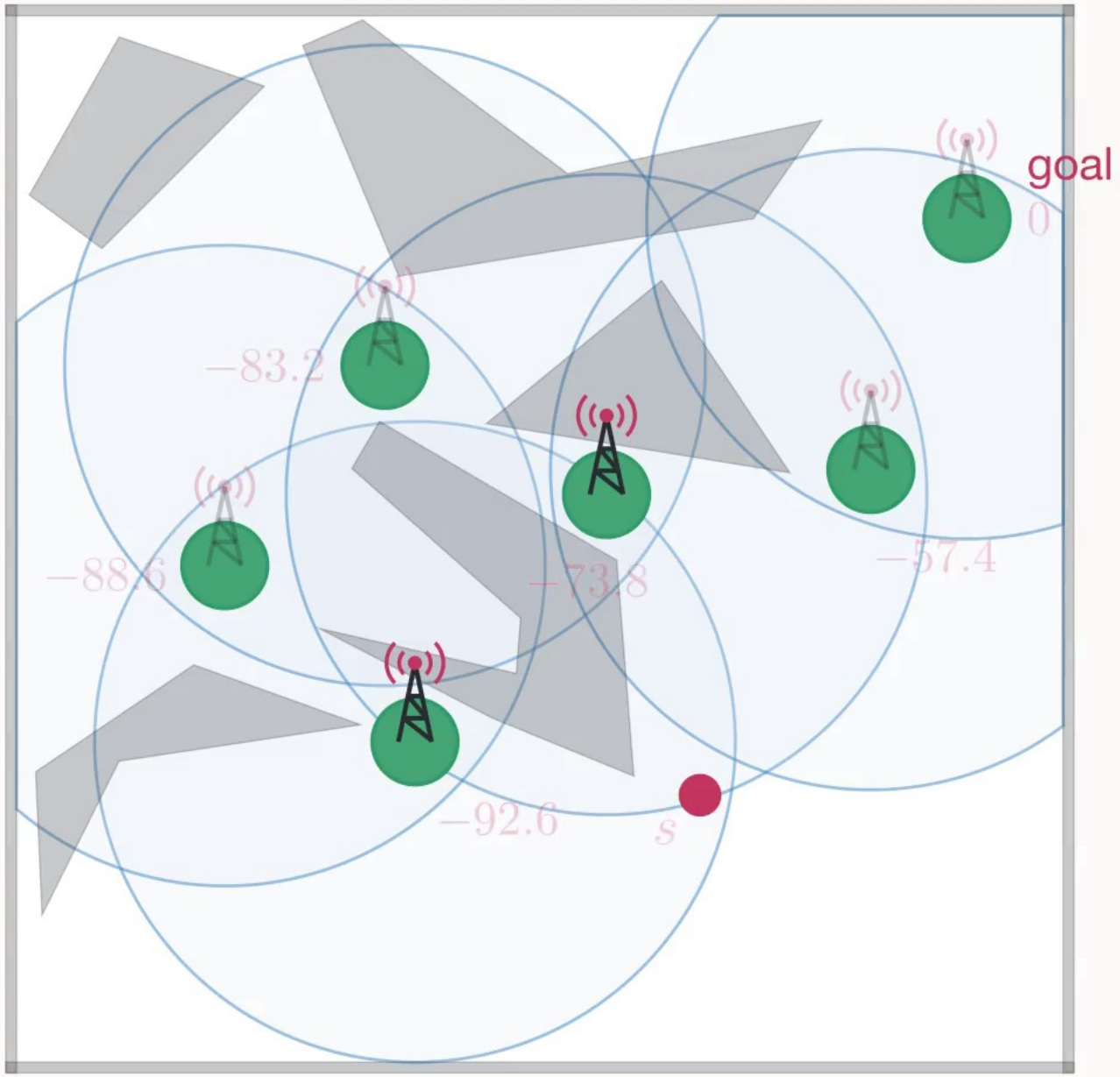
shaping gives the agent a more dense reward signal



Subgoals as radio towers

each subgoal broadcasts its value to the states in its initiation set

what would happen if the agent followed that option, not what to do



Subgoals as radio towers

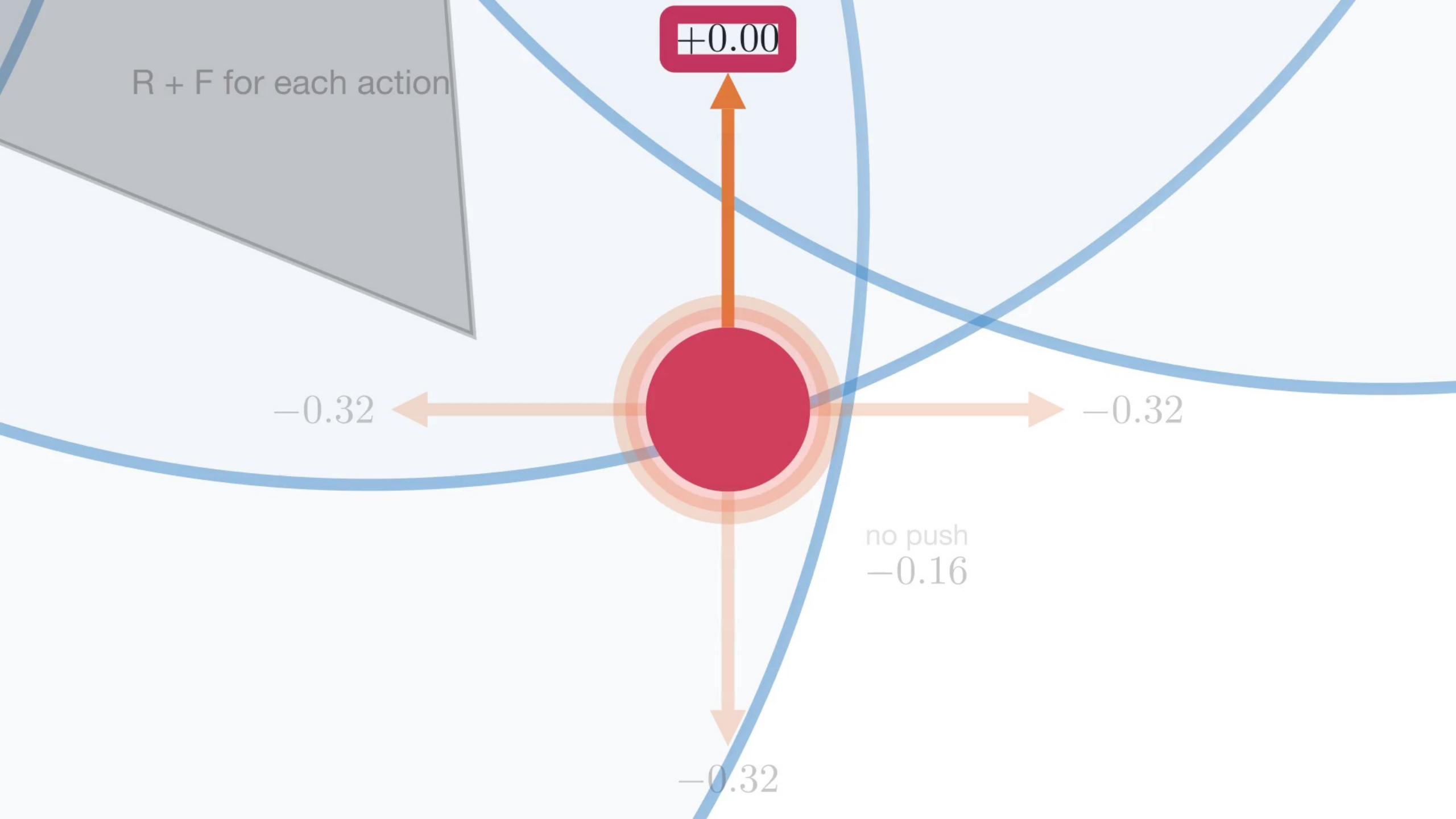
each subgoal broadcasts its value to the states in its initiation set

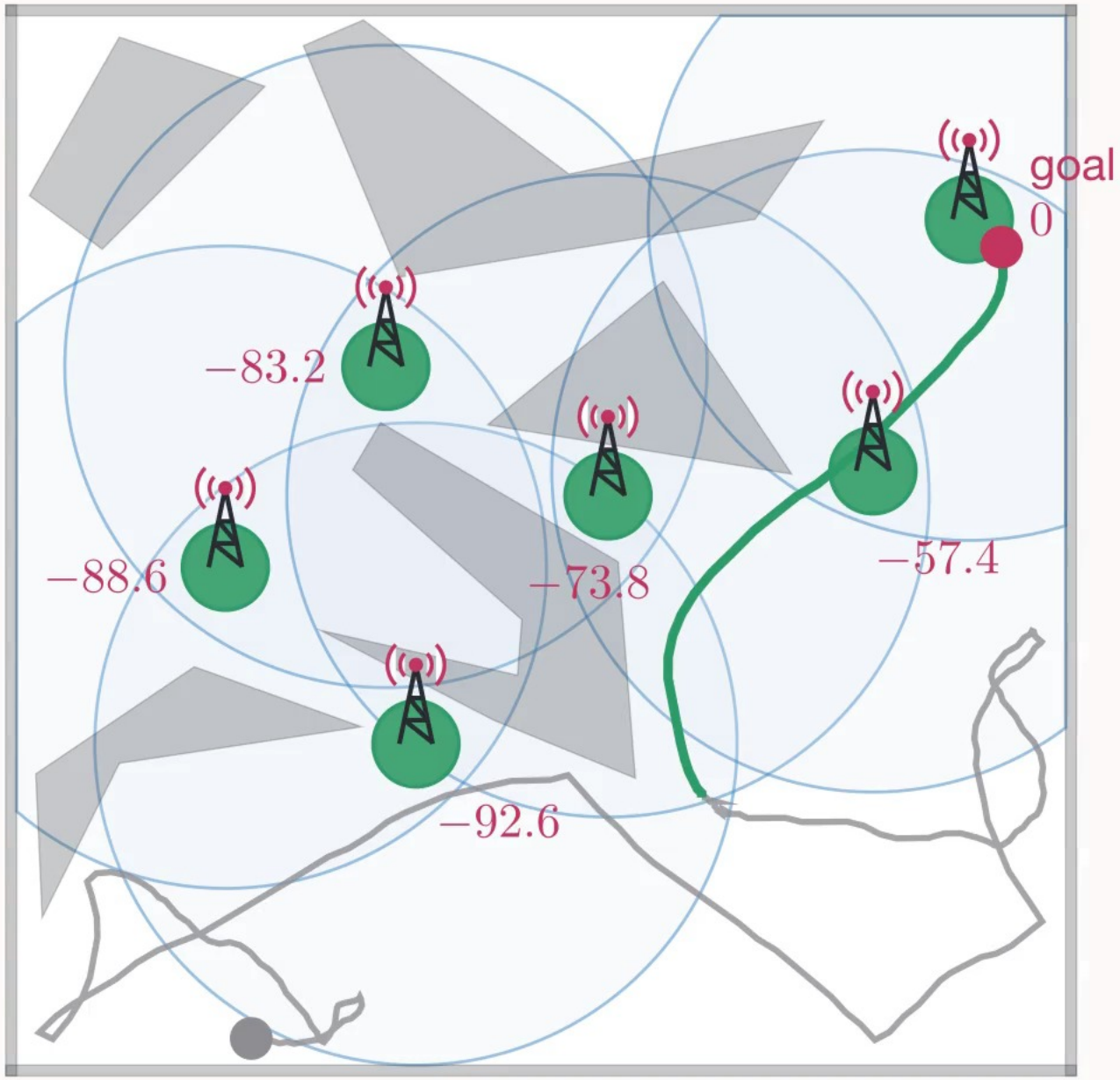
received at s:

$$r + \Gamma v^* = -40.1 + 0.60 \cdot (-92.6) = -95.57$$

$$r + \Gamma v^* = -38.9 + 0.61 \cdot (-73.8) = -84.00$$

$$v_p(s) = -84.00$$


R + F for each action
+0.00
−0.32
−0.32
no push
−0.16
−0.32

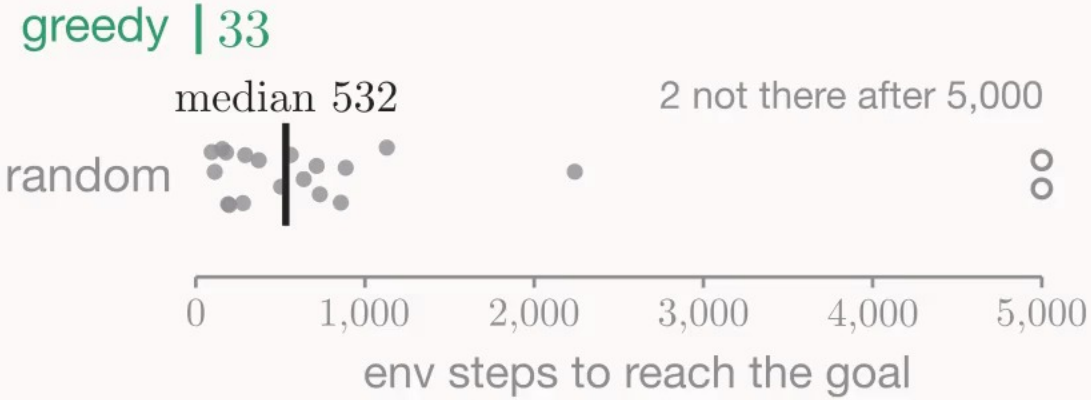


Subgoals as radio towers

greedy on R + F: **33 steps**

random: 200 steps, not there yet

20 random runs vs the greedy agent



about 16× fewer steps than random (median)

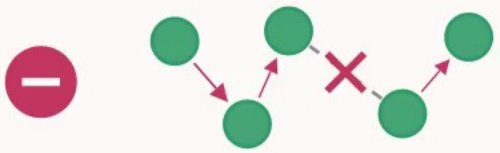
Conclusions & tradeoffs



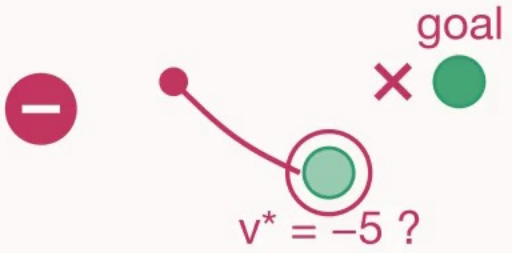
GSP tells the agent what would happen if it followed an option



it shortens the planning horizon and helps with long-term credit assignment

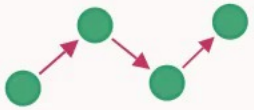


but it needs options with a specific structure (e.g. a connected subgoal graph)



and inaccurate models can really harm the agent

GSP vs the Option Keyboard

	 GSP	 Option Keyboard
options	needs a set of options with a specific structure	needs a good set of options with good features
learns	2 scalar-valued UVFAs r(s, g) and Γ(s, g)	a vector-valued GVF (successor features)
uses them to	do VI in the abstract MDP, then project values back as PBRS	combine the options into new ones (GPE + GPI)

Discussion

1. If all of this is so hard, is it worth using options at all?
2. Where do the options come from?
3. How do we handle inaccurate models?